

IMPLEMENTATION OF TOTAL QUALITY MANAGEMENT (TQM) IN HEALTHCARE

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ABSTRACT

Health services in India present a multiprovider scenario where the private and voluntary sectors play a key role in addition to the government sector. The word quality has many different definitions, ranging from conventional to those that are strategic. Services are defined as “social act(s) which take place in direct contact between customer and representatives of the service company. Service quality is more difficult to measure objectively than product quality because service characteristics include intangibility, heterogeneity, and inseparability of the production and consumption of services. The inaugural quality assurance work of Donabedian (1980) identified the importance of patient satisfaction as well as providing much of the basis for research in the area of quality assurance in healthcare. The important goals of hospitals are to deliver high quality health services and to respond to the needs of the patients. Patient satisfaction is one of the most sensitive indicators of the quality of their services. The concept of patient satisfaction has encouraged the adoption of marketing culture in service sector including health care services. This paper tries to present the dimensions of quality in healthcare, explain the relationship between health care and quality. Finally, this paper concludes the with the Total Quality Management (TQM) implementation in service hospitals.

Keywords: Quality, Healthcare, Critical Success Factor & TQM Implementation

JEL Classification Code: L0M0

Introduction

W. Edwards Deming, the father of concept Total Quality Management (TQM) has defines as a strategy aimed at the needs of the customer, present and future, whereas another quality guru, Joseph M. Juran has defined quality as conformance to requirements. Five broad attributes have been identified for understanding quality:

- Reliability – Ability to perform the promised service dependably and accurately.

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- Assurance – Knowledge and courtesy of employees.
- Tangibles – Physical facility, equipment and appearance of personnel.
- Empathy – Caring, individualized attention provided to the customers.
- Responsiveness – Willingness to help customers and provide prompt services.

Hospitals are by nature complex organizations, mainly because of its day long continuity of operation, emotional character of its services and intangibility of outcome, dual or even multiple lines of authority, high potential for conflict because of its highly divergent skill-mix and high labour intensive character. The complexity is compounded in service hospitals because of its service character, centralized control and a captive clientele with notions of perceived deficiencies (Chakravarthy et al. 2001).

Quality is considered a key factor in differentiation and excellence of services and is a potential source of sustainable competitive advantage so that its understanding, measurement, and improvement are important challenges for all health services organizations. Hospitals provide similar services with different quality. The quality can be used as a strategic differentiation for establishing a distinctive advantage, those difficult for rivals to follow or copy (Zarei, 2012). Most patients do not feel qualified to judge technical quality but assess their health care by other dimensions that reflect what they personally value (Kenagy et al, 1999).

Dimensions of Quality in Health Care

Quality is recognized as a multidimensional construct. Garvin developed a list of 8 dimensions of product quality. Garvin and others suggests that these dimensions are applicable to both products and services. However, difficulties arise when one tries to operationalize these dimensions in the service sector because service characteristics differ from product characteristics. Table 1 presents the dimensions of quality (Sower & et al., 2001).

Table 1: Dimensions of Quality

Garvin's Product Dimensions	Evans & Lindsay's Service Dimensions	Parasuraman et al.'s SERQUAL Dimensions	Chakrapani's Dimensions	Coddington's Dimensions	JACHO Dimensions
Performance Features Reliability Conformance Durability Serviceability Aesthetics Perceived Quality	Time Timeliness Completeness Courtesy Consistency Accessibility & Convenience Accuracy Response	Tangibles Reliability Responsiveness Assurance Empathy	Service/Product Depend-ability/ Support Exceeding Expectation	Warmth/Caring/ Concern Medical Staff Technology-- Equipment Specialization/ Services Available Outcome	Efficacy Appropriateness Efficiency Respect & Caring Safety Continuity Effectiveness Timeliness Availability

Source: Sower & et al., 2001

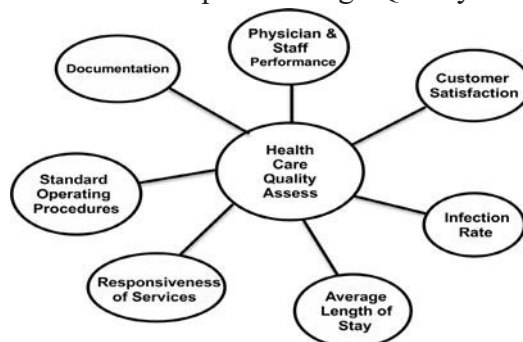
In the healthcare sector, the importance of measuring patient satisfaction is well articulated with patient satisfaction having been studied and measured extensively as a standalone construct and in particular in quality care assessment studies (Adeoti et al, 2012).

The works of Donabedian, Berwick, Blumenthal, Kazandjian, McLaughlin, Kalunzy and others are of great interest to healthcare quality management. Øvretveit (1997) has said that a new way of thinking in terms of evidence based medicine, health technology assessment, customer orientation and cost effective services are on the quality agenda of every healthcare professional. TQM is a new approach to quality improvement encompassing all systems and processes, clinical and non-clinical, with actions directed towards processes to improve the quality of all services and products for customers (Manjunath, 2010).

Relationship between Quality and Health Care

Health care is the diagnosis, treatment and prevention of disease, illness, injury and other physical and mental impairments in human beings. Health care is delivered by practitioners in allied health, dentistry, midwifery, obstetrics, medicine, nursing, optometry, pharmacy, psychology and other care providers. It refers to the work done in providing primary care, secondary care and tertiary care as well as in public health. Quality is the standard of something as measured against other things of a similar kind. The Institute of Medicine (IOM) has defined the quality of health care as ‘the degree to which health services for individuals and populations increase the likelihood of desired health outcomes and are consistent with current professional knowledge’. The quality of technical care consists in the application of medical science and technology in a way that maximizes its benefits to health without correspondingly increasing its risks. There are various parameters or indicators to judge the quality of health care. Patient satisfaction is the most important parameter to judge the quality of health care service provided (Guptha & Rokade, 2016).

Figure 1: Parameters Helpful to Judge Quality in Health Care



Source: Khushboo Sabharwal Gupta1 Varsha Rokade1, 2016

Indian Healthcare – A Snapshot

In the last two decades, the health services sector has grown rapidly in India. It is expected to touch USD 280 billion by 2020. It comprises hospital services, diagnostic services, diagnostic products, medical devices, medical technology, e-health services, clinical trial services, and clinical research organisations. This sector is predominantly privatised with almost 75 to 80 per cent of hospitals being managed by the private sector (DGCI&S, 2017).

Table 2: Fact Sheet of Indian Health Care (2015)

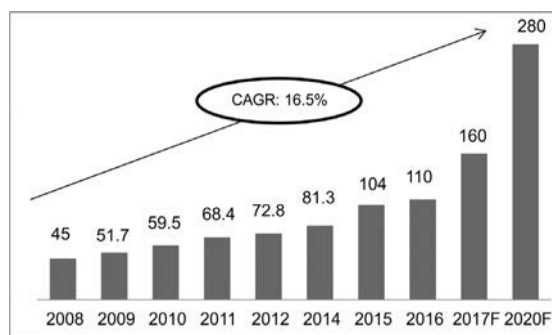
S. No.	Particulars	#
1.	Doctor : Patient Ratio (as per the MCI)	1:1,674
	Doctor : Patient Ratio (as per the WHO)	1:1,000
2.	Health Care Market size	USD110 Billion
3.	Hospital Beds per 1050 patients	1:1050
4.	Number of Sub Centres	156926
5.	Number of Hospitals	196312
6.	AYUSH Hospitals	3601
7.	Blood Banks	2760
8.	Organisations registered for Eye Banks	515
9.	Medical Colleges for MBBS (Government – 189 + Private – 215)	404
10.	Hospital Bed Patient Ratio	0.77083
11.	Doctors Density (per 1000 population)	1000
12.	Number of Post Graduate students per year	25346

Source: Indian Brand Equity Foundation (IBEF) Health Sector Presentation, January, 2017

Health Sector Growth Trend

Healthcare has become one of India's largest sectors both in terms of revenue and employment. The industry is growing at a tremendous pace owing to its strengthening coverage, services and increasing expenditure by public as well private players. During 2008-20, the market is expected to record a CAGR of 16.5 per cent. The total industry size is expected to touch USD160 billion by 2017 and USD280 billion by 2020 (IBEF, 2017).

Figure 2: Healthcare sector growth trend (USD billion)



Public Expenditure on Health

The three main categories of public expenditure on health care Medical & Public health, Family Welfare and others. Table 3 depicts the trends in public expenditure on health. During year 2009-10 Rs.72536 crores are expended on health, Rs.180656.77 cores are expended for Health during the year 2016 – 17 (BE). It is observed that per capita Expenditure on Health is shows the increasing trend (NHP – 2017).

Table 3: Trends in Public Expenditure on Health

Year	Public Exp. on Health (in Rs. crore)	Population (in crore)	GDP	Per Capita Public Expenditure on Health	Public Expenditure on Health as percentage of GDP (%)
2009-10	72536	117	6877827	321	1.12
2010-11	83101	118	7784115	701	1.07
2011-12	96221	120	8796039	802	1.10
2012-13	108236	122	9951344	890	1.09
2013-14	1102270	123	11272764	913	1.00
2014-15	121600.23	125	12433749	973	0.98
2015-16 (RE)	157743.37	126	13675331	1252	1.15
2016-17 (BE)	180656.77	128	15251028	1411	1.18

Source: National Health Profile – 2017 (p-187)

Table 4 depicts the number of Government Hospitals and Beds in Rural and Urban area in India, across various states.

Table 4: State/UT wise Number of Government Hospitals & Beds in Rural and Urban Areas (Including CHCs) in India (Provisional)

State/UT	Rural Hospitals		Urban Hospitals		Total Hospitals		Provisional/ Projected Population as on reference period in (000)	Average Population Served per Govt. Hospital	Average Population Served per Govt. Hospital Bed
	No.	Beds	No.	Beds	No.	Beds			
India	16816	183602	3490	492177	20306	675779	1238886	61011	1833
Andhra Pradesh	222	7380	56	12468	278	19848	86592	86952	438114
Arunachal Pradesh	204	2095	8	218	212	2313	1284	1284	555
Assam	1088	7504	49	5877	1137	13381	31693	31693	2369
Bihar	1325	5250	111	6302	1436	11552	101526	101526	8789
Chattisgarh	416	1522	221	10490	637	12012	25232	25232	2101
Goa	16	1684	15	1434	31	3118	1915	1915	614
Gujarath	296	8945	89	18983	385	27928	61329	61329	2196
Haryana	80	2454	79	5210	159	7664	26675	26675	3481
Himachal Pradesh	107	3328	53	5448	160	8776	6978	6978	795
Jammu & Kashmir	2368	5867	444	3893	2812	9760	12512	12152	1245

Jharkhand	545	4879	4	535	549	5414	32766	32766	6052
Karnataka	439	9884	215	43138	654	53022	61214	61214	1154
Kerala	1135	18082	143	20318	1278	38400	35258	35258	918
Madhya Pradesh	334	10020	117	18167	451	28187	75614	75614	2683
Maharashtra	450	12420	135	51445	585	63865	117189	117189	715
Manipur	23	730	7	697	30	1427	2534	2534	1776
Meghalaya	28	840	12	2287	40	3127	2712	2712	867
Mizoram	29	1420	7	210	36	1630	1039	1039	638
Nagaland	21	630	32	1797	53	2427	2327	2327	959
Odisha	1659	7099	91	9584	1750	16683	41797	41797	2505
Punjab	94	2900	146	8904	240	11804	28568	28568	2420
Rajasthan	2656	13038	489	13631	3145	26669	70969	70969	1521
Sikkim	30	730	3	830	33	1560	633	633	406
Tamil Nadu	407	9150	381	55093	788	64243	68654	68654	1069
Tripura	101	1022	21	3115	122	4137	3742	3742	904
Uttar Pradesh	737	NA	94	NA	831	0	211217	211217	NA
Uttarakhand	666	3746	29	4219	695	7965	10362	10362	1301
West Bengal	1272	19684	294	58882	1566	78566	91920	91920	1170
A & N Islands	31	625	1	450	32	1075	533	533	495
Chandigarh	0	0	4	700	4	700	1651	1651	2359
D & N Haveli	1	100	1	272	2	372	402	402	1080
Daman & Diu	0	0	4	200	4	200	305	305	1523
Delhi	0	0	109	24383	109	24383	20092	20092	824
Lakshadweep	9	300	0	0	9	300	78	78	261
Puducherry	27	274	26	2997	53	3271	1573	1573	48181

Source: National Health Profile, 2015, p.243.

Table 5 presents the key corporate players in the Indian Health Care market with their bed capacity and presence.

Table 5: Key Players in the Market (Private)

S.No.	Name of the Player	Beds	Presence
1.	Apollo Hospitals Enterprises Limited	9215	32 cities
2.	Aravind Eye Hospitals	3649	12 cities
3.	Care Hospitals	2100	10 cities
4.	Fortis Health Care Limited	10000	16 cities
5.	Max Hospitals	2016	04 cities
6.	Manipal Group of Hospitals	4900	10 cities
7.	Narayana Health	7452	19 cities

Source: IBEF, Health Care – 2017.

Vast opportunities for investment in healthcare infrastructure in both urban and rural India. About 1.8 million beds required by the end of 2025. Additional 1.54 million doctors

and 2.4 million nurses required to meet the growing demand. Table 6 depicts comparison of Health Infrastructure in India with WHO standards.

Table 6: Health Infrastructure per 1,000 Individuals (2013 – 2015)

	Physicians	Nurses & Midwifery Personnel	Hospital Beds
India	0.7	1.5	1
World Median	2.5	2.5	2.9

Source: IBEF, 2017

Importance of Quality in Health Care

In recent years, the role of governments as the main provider of health-care services has changed. Initially, governments mainly provided funding, but now they increasingly focus on issues of organization, management and quality in hospitals. Governments have developed various initiatives to promote and improve quality management. Additionally, the health-care market itself is changing from a seller-oriented to a buyer oriented market due to the increasing influence of consumer organizations, insurance companies and public pressure. As a consequence, the patient is becoming a customer (Kunst & Lemmink, 2000).

Health services include a wide variety of quality aspects, all of which are important. In the case of medical services, the seller is doctors, hospitals, nursing homes, clinics, etc. because they offer health services for sale at stipulated prices. The buyer is the client or patient who buys these health services at the stipulated prices. It may also include quality of performance that is directly connected and closely related to healthcare such as food, housing, safety, security, attitude of employees, and other factors that arise in connection with hospitals and nursing homes. So, the time taken to fix an appointment, delay time, services time, timing with regard to medical treatment and surgery.

- Quality of administration and management
- Quality of doctors
- Quality of hospital care (Patel).

Quality has become an increasingly important means of competition in the healthcare environment. Patients are becoming more demanding consumers. Hospitals, like all service companies have problems in defining quality, providing and controlling the desired levels of performance.

- Product based - quality is defined as precise and measurable
- User based - quality is defined as fitness for intended use
- Value based - quality is defined in terms of costs and price

The definitions put forward by Crosby and Juran were found to be widely accepted, that is 'quality is fitness for use' (Juran, 1988) and 'quality is meeting requirements' (Crosby, 1979).

1. To the medical staff - 'quality is about "curing", effectiveness of medical care, thoroughness and clinical outcomes'.
2. To the receptionists - 'quality is about how we attend to patients queries'.
3. To the Chief Executive - 'quality is low cost'.
4. To the Quality Manager - 'quality is about meeting every patients needs if possible (Lim & Tang).

Implementation of Quality Management in Healthcare

R.P. Mohanty & etal identified a number of factors that can influence the implementation of TQM. Broadly, they may be classified as:

- Organizational factors;
- Interpersonal factors;
- Environmental factors;
- Facilities factors; and
- Economic factors.

Organizational factors Organizational factors are the basic components of any health-care system – the staff resources, in terms of number, variety and level. Some of these resources are internal – the staff actually employed within the particular organization, and some are external – additional staff resources which may have to be used in case of a shortage of the primary, internal resource.

Interpersonal factors All systems are influenced by interpersonal factors. This is evidently so in HCS, where quality can be significantly influenced by individual exchanges between administrators, board members, physicians, nurses and other medical personnel, as well as the way in which those parties interact with patients.

Environmental factors All open systems are influenced by their environment – factors which normally do not lie within the control of the system. Like other systems, the environment of this system is dynamic. A knowledge of these factors is necessary to predict the behaviour of the system as these external variables change over time.

Facilities factors Facilities refer to the non-staff resources of the health-care system. These will be medical, technical and ancillary (the quality of healthcare as perceived by a patient may be influenced strongly by non-medical facilities such as catering).

Economic factors Health-care systems have to be economically viable. This, as in any other system, is determined by the balance between the availability of funds and the costs of services. Of course, it also depends on how funds are deployed. A long-term perspective may result in immediate investment in new technology to save costs in the longer term(RP Mohanthy et al, 1996).

Critical Success factors Influencing TQM

TQM adoption and implementation requires changes in structure, system, and process as a necessary precondition to achieve improved business performance and changes in employee behavior. It is therefore important to identify the critical factors that influence the success of TQM adoption and implementation in service organizations. Some of these critical factors are briefly discussed.

- Factor – I: Top Management Commitment
- Factor – II: Employee Involvement
- Factor – III: Customer Satisfaction
- Factor – IV : Team Work
- Factor – V: Processes
- Factor – VI: Continuous Improvement
- Factor – VII: Training
- Factor – VIII: Culture Change
- Factor – I Top Management Commitment

Management acts as the driver for TQM implementation, creating values, goals and systems to satisfy customer expectations and to improve an organization's performance and responsible for providing direction and encouragement to the organization. Management commitment is crucial for a company's quality development since, with their support and contributions, adequate resources will be allocated to enhance the training activities resulting in better quality measurement, improved customer satisfaction and benchmarking.

Factor – II: Employee Involvement

Employee involvement is a critical component of TQM. TQM requires total management commitment to ensure employees indulge in quality work culture and hence create healthy corporate image by rendering quality services to the customers.

Factor – III: Customer Satisfaction

TQM makes customer satisfaction the number one organization priority, where an emphasis is placed on meeting or exceeding external customer expectations in every transaction.

Factor - IV: Teamwork:

Teamwork is a critical factor in TQM as teamwork is essential in having a fully functioning process management and improvement, especially in medical treatment; it requires cooperation among all related departments.

Factor – V: Processes

TQM is centered for an effective management of processes and continuous customer satisfaction. The process is improved by reducing the source of variation that exists within it and everyone, in the TQM environment, is required to gain additional capabilities to improve the process.

Factor – VI: Continuous Improvement

Continuous improvement is a powerful concept related to the pursuit of never-ending improvement in meeting external and internal customer needs.

Factor – VII: Training

Training is a very important tool for promoting and developing skills related to an organization's beliefs and values to change to a culture that places high value on quality. Once management has the skills to lead the TQM process, the rest of the organization should be trained to ensure a systematic, integrated, consistent organization-wide effort.

Factor – VIII: Culture Change

Instilling quality-oriented culture requires change of attitudes, value systems, and beliefs. TQM is an educational process aiming at changing the behavior and attitudes of organizational members and then developing quality sensitive organizational culture (Munhurrin, 2011).

Outline for TQM implementation in Service Hospitals

It will have to be appreciated before planning implementation of TQM that the process of total quality management is a long term strategic initiative which requires adequate time for implementation. With the Organisational culture. General sequence of typical Quality Management Process in service hospitals will be as follows:

1. Designing the quality improvement organizational structure
2. Quality Management Awareness and Training
3. Quality Management Framework Development
4. Quality management practice
5. Customer Awareness development
6. Organisational Quality Awareness Building

7. Ongoing training and cross-organisational outcome measure
8. Project dominant focused efforts

Conclusion

Improving the quality of health care at the system level requires a focus on governance issues, including improving public-sector management, building institutional capacity, and promoting a culture of data-driven policies. Setting performance standards and expectations is another essential element to improving patient safety. From a sociotechnical point of view a quality system is an association, a network of humans and artefacts, formed with a view to some good purpose, e.g. improving the quality of medical practice by internal reflexivity of medical professionals in health care organisations and (a valid) external accountability to patients or society.

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MJ SSIM IX (II), 4, 2017